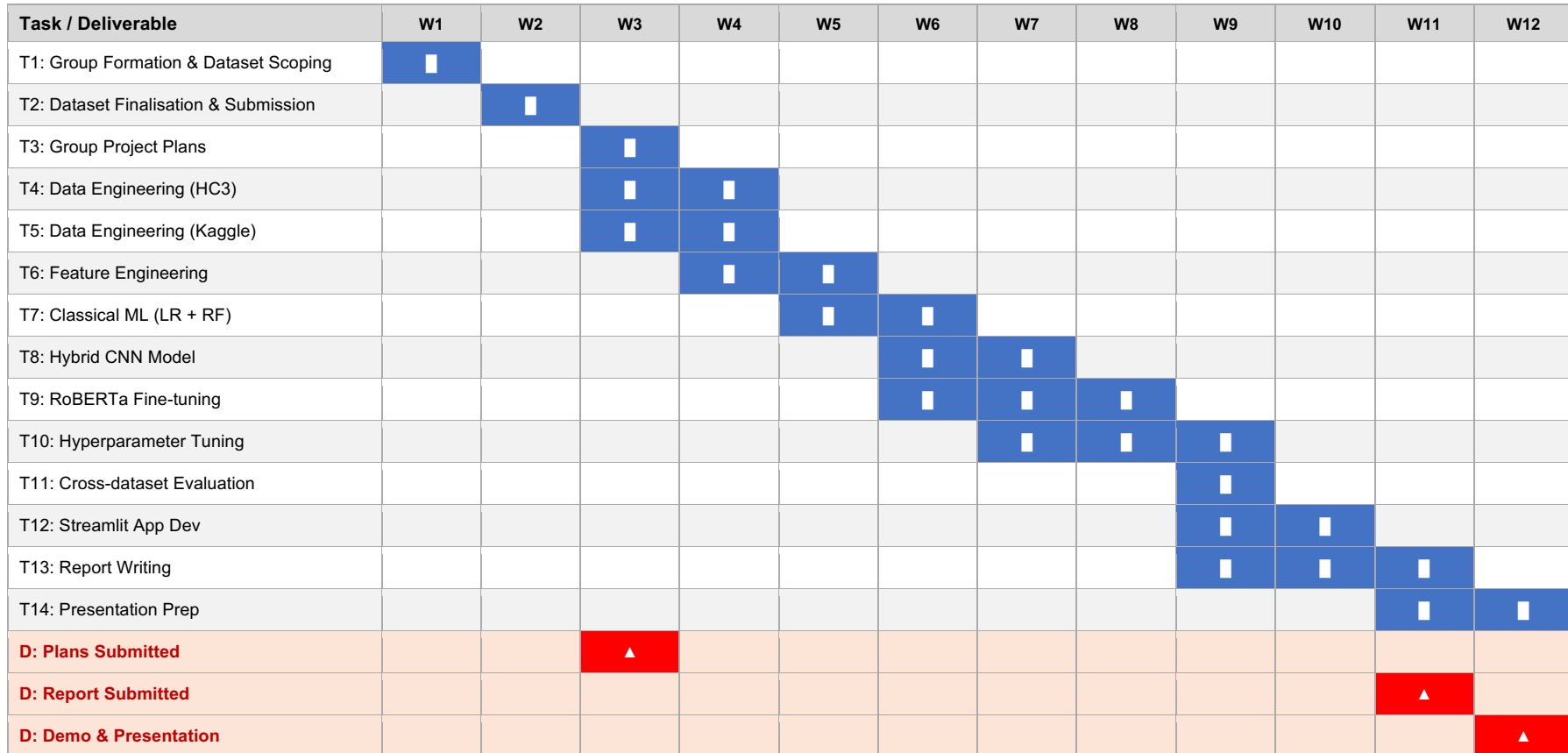


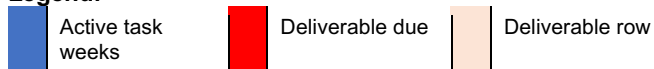
Project Timeline

Detecting AI-Generated Text Using ML and NLP Techniques

2.1 Gantt Chart (12 Weeks)



Legend:



2.2 Task Detail Schedule

Task	Description	Start	End	Owner(s)
T1	Group Formation & Dataset Scoping	Wk 1	Wk 1	All
T2	Dataset Finalisation & Submission	Wk 2	Wk 2	All
T3	Group Project Plans (6 docs to Canvas + GitHub)	Wk 3	Wk 3	All
T4	HC3 Engineering: load, flatten, label, EDA	Wk 3	Wk 4	P1
T5	Kaggle Engineering: CSV load, schema align, merge, splits	Wk 3	Wk 4	P2
T6	Feature Engineering: TF-IDF, stylometrics, GloVe	Wk 4	Wk 5	P3 & P4
T7	Classical ML: Logistic Regression, Random Forest, baselines	Wk 5	Wk 6	P3
T8	Hybrid CNN: dual-stream architecture + GloVe training	Wk 6	Wk 7	P4
T9	RoBERTa Fine-tuning via HuggingFace Trainer + early stop	Wk 6	Wk 8	P5 & P6
T10	Hyperparameter Tuning (GridSearch, manual grid, sweep)	Wk 7	Wk 9	P3–P6
T11	Cross-dataset Evaluation (HC3 ↔ Kaggle)	Wk 9	Wk 9	All
T12	Streamlit App: UI, model integration, validation, deploy	Wk 9	Wk 10	P5 & P6
T13	Full Academic Report (figures, discussion, references)	Wk 9	Wk 11	All
T14	Presentation Prep: slides, live demo, dry run	Wk 11	Wk 12	All
D	Plans Submitted to Canvas + GitHub	Wk 3	Wk 3	All
D	Final Report Submitted	Wk 11	Wk 11	All
D	Live Demo & Group Presentation	Wk 12	Wk 12	All

2.3 Key Milestones

Wk 1, Day 5: Group formed; topic and datasets agreed. GitHub repo created.

End of Wk 2: Datasets finalised. Kaggle CSV downloaded; HC3 loadable programmatically.

End of Wk 3: All six planning documents submitted on Canvas and committed to GitHub.

End of Wk 4: Cleaned merged corpus (train.csv / val.csv / test.csv) committed. EDA reviewed.

End of Wk 6: All three models implemented and meeting minimum KPI thresholds on val set.

End of Wk 8: Hyperparameter tuning complete. Cross-dataset evaluation done. Best checkpoint saved.

End of Wk 10: Streamlit app running locally, passing all UAT scenarios.

End of Wk 11: Final report submitted. Code merged to main and tagged v1.0. Repo made public.

Week 12: Live app demo delivered. All team members participate in Q&A.

2.4 Risk & Contingency

Risk	Likelihood	Mitigation
Team member becomes unavailable	Medium	All tasks in GitHub Issues. 48-h reassignment protocol.
Colab GPU quota exhausted	High	P5 monitors weekly. Fallback: frozen RoBERTa (~88% F1).
GitHub merge conflicts	Medium	Clear branch strategy; P6 resolves within 24 h.
Report writing bottleneck	Medium	Writing starts Wk 9 in parallel with modelling.
Kaggle dataset access failure	Low	SHA-256 checksum + Google Drive backup; HC3 as fallback.